

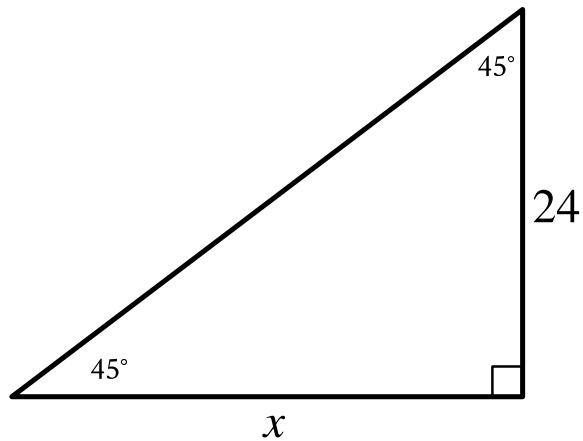
● EASY

● GEOMETRY AND TRIGONOMETRY

Right triangles and trigonometry

01

10 questions · ~15 min

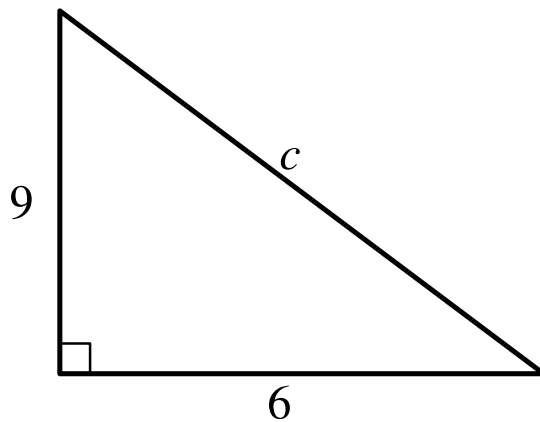


Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of the other 2 angles are as follows:
 - 45°
 - 45°
- The lengths of the 2 sides adjacent to the right angle are as follows:
 - x
 - 24
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

- A. 24
- B. 45
- C. 48
- D. 69



Note: Figure not drawn to scale.

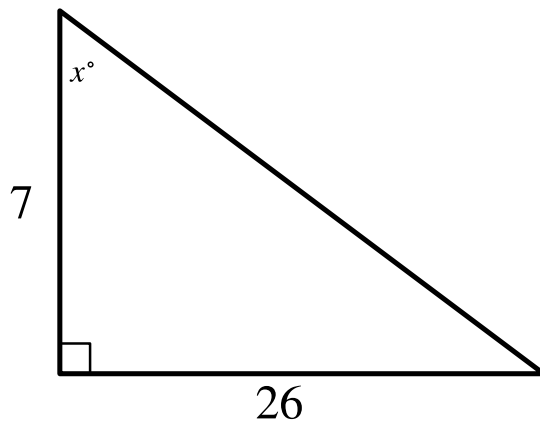
- One angle is a right angle.
- The length of the side opposite the right angle is c .
- The lengths of the other 2 sides are as follows:
 - 9
 - 6
- A note indicates the figure is not drawn to scale.

In the right triangle shown, which of the following is closest to the value of c ?

- A. 7.5
- B. 10.8
- C. 15
- D. 58.5

A right triangle has legs with lengths of 11 centimeters and 9 centimeters. What is the length of this triangle's hypotenuse, in centimeters?

- A. $\sqrt{40}$
- B. $\sqrt{202}$
- C. 20
- D. 202

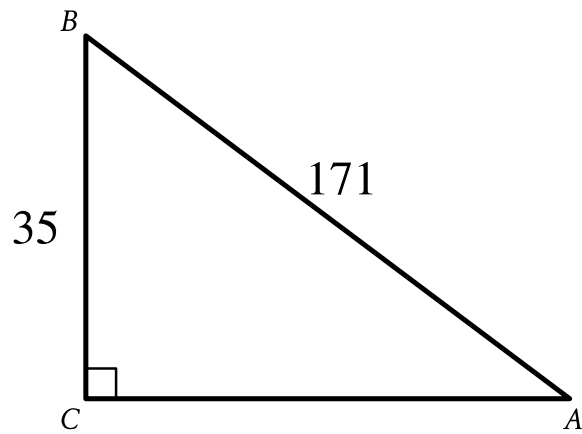


Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The length of the leg opposite the angle with measure x° is 26.
- The length of the leg adjacent to the angle with measure x° is 7.
- A note indicates the figure is not drawn to scale.

In the triangle shown, what is the value of $\tan x^\circ$?

- A. $\frac{1}{26}$
- B. $\frac{19}{26}$
- C. $\frac{26}{7}$
- D. $\frac{33}{7}$

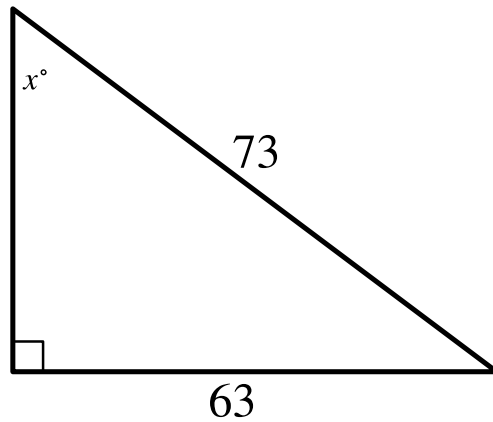


Note: Figure not drawn to scale.

- Angle upper C is a right angle.
- The length of side upper A upper B is 171.
- The length of side upper B upper C is 35.
- A note indicates the figure is not drawn to scale.

In the right triangle shown, what is the value of $\sin A$?

- A. $\frac{1}{171}$
- B. $\frac{35}{171}$
- C. $\frac{171}{35}$
- D. 171

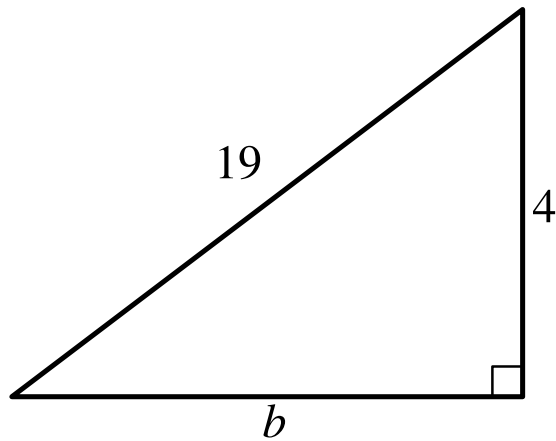


Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The length of the side opposite the right angle is 73.
- The length of the side opposite the angle with measure x° is 63.
- A note indicates the figure is not drawn to scale.

In the right triangle shown, what is the value of $\sin x^\circ$?

- A. $\frac{1}{73}$
- B. $\frac{10}{73}$
- C. $\frac{63}{73}$
- D. $\frac{136}{73}$



Note: Figure not drawn to scale.

- One angle is a right angle.
- The length of the side opposite the right angle is 19.
- The lengths of the other 2 sides are as follows:
 - b
 - 4
- A note indicates the figure is not drawn to scale.

Which equation shows the relationship between the side lengths of the given triangle?

- A. $4b = 19$
- B. $4 + b = 19$
- C. $4^2 + b^2 = 19^2$
- D. $4^2 - b^2 = 19^2$

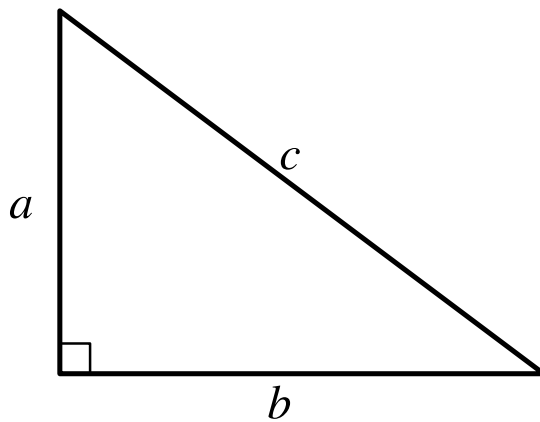
Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D , and angles C and F are right angles. If $\cos B = \frac{1}{22}$, what is the value of $\cos E$?

A. $\frac{1}{22}$

B. $\frac{1}{23}$

C. $\frac{21}{22}$

D. $\frac{22}{23}$

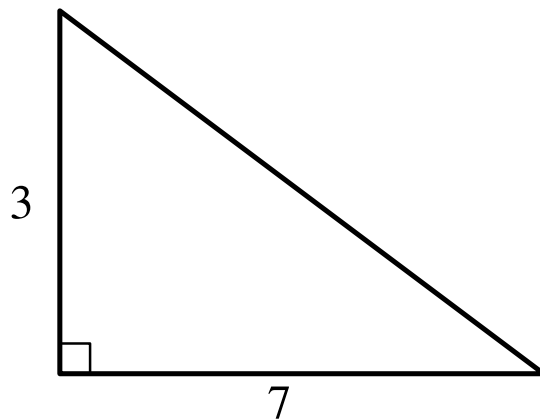


Note: Figure not drawn to scale.

- The side lengths are labeled as follows:
 - a
 - b
 - c
- The angle opposite the side labeled c is a right angle.
- A note indicates the figure is not drawn to scale.

For the right triangle shown, $a = 4$ and $b = 5$. Which expression represents the value of c ?

- A. $4 + 5$
- B. $\sqrt{45}$
- C. $\sqrt{4 + 5}$
- D. $\sqrt{4^2 + 5^2}$



Note: Figure not drawn to scale.

- One angle is a right angle.
- The lengths of the sides adjacent to the right angle are as follows:
 - 3
 - 7
- A note indicates the figure is not drawn to scale.

The lengths of the legs of a right triangle are shown. Which of the following is closest to the length of the triangle's hypotenuse?

- A. 3.2
- B. 5
- C. 7.6
- D. 20